

TrashUQ: Federated Learning for Intelligent Waste Monitoring on Arduino UNO Q

Aleix BERTRAN ANDREU^a Pol LLINÀS VAQUER^a Bru PALLÀS VARGUÉS^a
Aleix ROSINACH OLIVART^a

^a *Universitat de Lleida, Lleida, Catalonia*

Abstract. This paper presents TrashUQ, a waste-monitoring platform that integrates Arduino UNO Q edge nodes, MQTT telemetry, and a gRPC Federated Learning (FL) coordinator. We report two validations. First, a real two-device deployment confirms end-to-end telemetry and FL round-trip behavior (282 persisted MQTT records, model version 0→26, 25/30 successful aggregations). Second, a controlled non-IID FL simulation (Dirichlet $\alpha = 0.3$) with 2, 5, 10, and 20 clients shows stable FedAvg convergence over 25 rounds, with final accuracies around 93–94%. We further demonstrate that stochastic rounding gradient quantization achieves 4–8× model compression with negligible accuracy degradation (0–1.33%). Results indicate that FL orchestration is feasible on this embedded stack, and that gradient quantization effectively mitigates the communication bottleneck.

Keywords. Federated Learning, Edge AI, Waste Classification, Arduino UNO Q, MQTT, TrashNet

1. Introduction

Waste mismanagement poses critical environmental and public health challenges. Intelligent waste sorting systems powered by deep learning can significantly improve recycling rates, but conventional approaches rely on centralized cloud infrastructure that raises privacy, latency, and bandwidth concerns [1]. Federated Learning (FL) addresses these limitations by training models collaboratively across edge devices while keeping data local [2,3].

Several compression strategies have been proposed to mitigate the communication bottleneck in FL, including stochastic quantization (QSGD [4]), 1-bit sign compression (SignSGD [5]), top-k sparsification [6], and unified compressed FL frameworks (FedCOM [7]). Hierarchical FL [8,9] further reduces coordinator load through multi-level aggregation. However, most prior work evaluates these techniques exclusively in simulation with high-end hardware, leaving open questions about their applicability to resource-constrained edge deployments. The Arduino UNO Q—featuring a Qualcomm QRB2210 processor—enables practical FL deployment on physical low-cost devices.

This paper presents TrashUQ, a complete FL-based waste monitoring system deployed on two Arduino UNO Q boards. Our contributions are:

1. A real two-node FL deployment with MQTT telemetry validation, confirming end-to-end FL round-trip behavior (282 records, 25/30 successful aggregations).
2. FL simulation results on a controlled synthetic dataset with FedAvg across 2–20 non-IID clients, demonstrating stable convergence around 93–94% accuracy.
3. Stochastic rounding gradient quantization achieving $4\text{--}8\times$ model compression with negligible accuracy degradation (0–1.33%).
4. Open-source implementation at <https://github.com/TrashUQ/TrashUQ> [10].

2. System Architecture

Each Arduino UNO Q has a Qualcomm QRB2210 processor (Debian Linux) and an STM32U585 microcontroller, connected over Wi-Fi to a shared backend (MQTT broker, PostgreSQL, FastAPI REST API, gRPC FL coordinator). The edge daemon follows capture/classify/act logic using a MobileNetV2 TFLite model (classes: cardboard, glass, paper, plastic). User corrections are stored locally for on-device fine-tuning.

The trainable model is a frozen MobileNetV2 backbone with a small learnable head. After accumulating new labels, a node fetches global weights, trains locally, and submits an update via gRPC; the coordinator applies FedAvg [1]. Boards publish status and FL metrics to `arduino/<device-id>/#` via MQTT.

3. Hardware Validation

Two Arduino UNO Q nodes (Debian 13) ran the edge daemon against the shared backend during a ~ 12 -minute session. End-to-end ingestion was validated across Edge $\rightarrow$ MQTT $\rightarrow$ Backend $\rightarrow$ PostgreSQL $\rightarrow$ Frontend, persisting 282 MQTT rows (147 from `unoq-01`, 135 from `unoq-02`).

Both nodes successfully completed FL cycles (`Join`, `GetGlobalModel`, `SubmitUpdate`), reaching model version 26 with 25/30 successful aggregations (83.3%). Five submissions were rejected as stale during concurrent updates, confirming correct monotonic round enforcement. Round submission latency was low (median ~ 25 ms, max 105 ms). Mean local accuracy was 0.921 (`unoq-01`) and 0.871 (`unoq-02`). Both devices used `-fake-camera` and HTTP label injection to exercise FL deterministically; camera-in-the-loop latency is left for future work.

4. FL Simulation and Communication Efficiency

We ran FedAvg for 25 rounds with non-IID partitions (Dirichlet $\alpha = 0.3$), using 3 seeds per setting with client counts $\{2, 5, 10, 20\}$. Each run used local epochs = 2, batch size = 16, SGD with learning rate 0.18, and full participation.

To enable reproducible experiments without external dependencies, we generated a synthetic dataset of 16×16 grayscale images (4 waste classes, 220 samples/class). Each class prototype—horizontal stripes (cardboard), diagonal lines (glass), nested rectangles (paper), vertical stripes (plastic)—is blended via convex combination (52% main, 23% next, 10% alternate, 15% shared) with Gaussian noise ($\sigma = 0.30$), brightness jitter, and pixel roll. The simulator can optionally load real TrashNet images if available.

Table 1. FL simulation summary (mean across 3 seeds).

Clients	Acc. (%)	Loss	Round Time (s)	Comm. (MB)
2	93.75	0.2192	0.005	0.394
5	93.56	0.2004	0.005	0.985
10	93.37	0.2018	0.006	1.970
20	93.75	0.1933	0.008	3.941

Table 2. Quantization results (20 clients, mean across 3 seeds).

Config	Acc. (%)	Loss	Comm. (MB)	Compression
float32	93.75	0.1933	3.941	1.0×
8-bit	93.75	0.1935	2.474	~4×
4-bit	93.75	0.1976	2.229	~8×

4.1. Convergence and Communication Efficiency

Across all settings, FL converged stably with final accuracy near 93–94% and no failed rounds (Table 1). Mean round duration increased from 0.005 s (2 clients) to 0.008 s (20 clients), while total communication grew approximately linearly from 0.394 MB to 3.941 MB.

4.2. Gradient Quantization

To address the communication bottleneck, we integrate stochastic rounding quantization into the FedAvg pipeline. After local training, each client quantizes weight updates before transmission; the server dequantizes before weighted averaging. We evaluate 8-bit ($4\times$ weight compression) and 4-bit with uint8 packing ($8\times$ weight compression).

The 8-bit configuration preserves accuracy identically (93.75%) with $\sim 4\times$ model compression. The 4-bit packed variant achieves $\sim 8\times$ compression with a maximum accuracy degradation of 1.33% (at 2 clients), diminishing to zero as the client count increases. These results demonstrate that lightweight gradient quantization effectively mitigates the communication bottleneck without compromising model quality on resource-constrained edge hardware.

5. Conclusion and Future Work

TrashUQ validates FL on real Arduino UNO Q hardware (two-node deployment, 282 MQTT records, 25/30 successful FL aggregations) and in controlled simulation (FedAvg, 2–20 non-IID clients, 93–94% accuracy). Stochastic rounding gradient quantization achieves $4\text{--}8\times$ model compression with negligible accuracy loss (0–1.33%), effectively mitigating the communication bottleneck. Code at <https://github.com/TrashUQ/TrashUQ>; future work includes camera-in-the-loop validation and on-device quantized model deployment.

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